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ACID RADICALS (ANIONS)

What to do + why it works

Carbonate (CO_3^{2-})
What you do
Take a pinch of salt in a test tube
Add a few drops of dilute HCl
What you see
Bubbles come out fast
What you do next (to be sure)
Pass that gas into lime water
What happens
Lime water turns milky
Why this proves carbonate
Carbonate reacts with acid to give CO_2 gas.
CO_2 turns lime water milky. No other ion does this.
✓ Carbonate confirmed

Chloride (Cl^-)
What you do
Dissolve salt in water
Add dilute nitric acid
Add silver nitrate solution
What you see
White, curdy solid appears
What you do next (confirm)
Add ammonium hydroxide (NH_4OH)
What happens
White solid disappears
Why this proves chloride
Silver chloride dissolves in NH_4OH.
Other silver salts don’t behave like this.
✓ Chloride confirmed

Sulphate (SO_4^{2-})
What you do
Take salt solution
Add dilute HCl
Add barium chloride
What you see
White solid forms
What you do next
Add concentrated HCl
What happens
White solid does NOT dissolve
Why this proves sulphate
Barium sulphate is insoluble even in strong acid.
✓ Sulphate confirmed

Nitrate (NO_3^-)
What you do
Take salt solution
Add fresh $FeSO_4$ solution
Slowly pour conc. H_2SO_4 along the side
What you see
Brown ring forms in between layers
Why this proves nitrate
Nitrate makes a special brown complex at the layer.

✓ Nitrate confirmed

Acetate / Acetic acid (CH_3COO^- / CH_3COOH)

What you do

Take salt in test tube

Add ethanol

Add few drops conc. H_2SO_4

Warm gently

What you smell

Sweet fruity smell

Why this proves acetate

Acetate makes ester, and esters smell fruity.

✓ Acetate confirmed

Oxalate ($C_2O_4^{2-}$)

What you do

Take salt solution

Add acidified $KMnO_4$

Warm gently

What you see

Purple colour disappears

Why this proves oxalate

Oxalate destroys the purple colour of $KMnO_4$.

✓ Oxalate confirmed

BASIC RADICALS (CATIONS)

☁ Ammonium (NH_4^+)

What you do

Heat salt with NaOH

What you see

Gas comes out

What you do next

Bring HCl dipped glass rod

What happens

White fumes appear

Why this proves ammonium

Only ammonia gas gives white fumes with HCl.

✓ NH_4^+ confirmed

Lead (Pb^{2+})

What you do

Add dilute HCl to salt solution

What you see

White solid forms

What you do next

Heat the solution

What happens

Solid dissolves

Why this proves lead

Lead chloride dissolves in hot water.

✓ Pb^{2+} confirmed

Copper (Cu^{2+})

What you do

Add H_2S gas to acidic solution

What you see

Black solid forms

Why this proves copper

Copper sulphide is black.

✓ Cu^{2+} confirmed

Aluminium (Al^{3+})

What you do
Add NaOH drop by drop
What you see
White jelly-like solid
What you do next
Add excess NaOH
What happens
Solid dissolves
Final check
Do ash test
Why this proves aluminium
Aluminium dissolves in excess NaOH and gives blue ash.
✓ Al^{3+} confirmed

Zinc (Zn^{2+})

What you do
Add NaOH slowly
What you see
White solid
What you do next
Add excess NaOH
What happens
Solid dissolves
Final check
Ash test
Why this proves zinc
Zinc gives green ash, different from aluminium.
✓ Zn^{2+} confirmed

Calcium / Strontium / Barium

What you do
Perform flame test
What you see
Brick red → Ca^{2+}
Crimson → Sr^{2+}
Pale green → Ba^{2+}
Why this proves them
Each metal burns with a unique colour.
✓ Ion confirmed